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SHORT-PAPER

## Short Paper: Predicting and Analyzing EV Energy Consumption in Bangladesh : A Machine Learning Approach

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# Short Paper: Predicting and Analyzing EV Energy Consumption in Bangladesh : A Machine Learning Approach

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## Abstract

The increasing adoption of EVs in Bangladesh poses challenges to the existing power grid [15]. This study develops a machine learning model to accurately predict EV energy consumption at charging stations [23], using California City data as a proxy [8]. Evaluating algorithms like Random Forest, Gradient Boosting Regression (GBR), and Decision Tree [3] [7], Gradient Boosting outperformed with an  $R^2$  of 0.9999965 and an MAE of 0.0071. Beyond predictions, this research optimizes energy management, assesses grid impacts, and supports sustainable infrastructure development, offering crucial insights for integrating EVs into Bangladesh's transportation sector.

## CCS Concepts

• Computing methodologies → Machine learning algorithms; Supervised learning by regression; Supervised learning by regression; • Applied computing → Multi-criterion optimization and decision-making.

## Keywords

Machine learning, Energy, EV charging, EV, Electric Power, Bangladesh

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## 1 Introduction

The global shift to cleaner energy has accelerated EV adoption, offering a sustainable solution to reduce transportation-related carbon emissions [15], [30]. While EVs contribute to decarbonization,

they also place significant strain on power networks, especially in developing countries like Bangladesh, where increased demand for EV charging burdens the national grid during peak hours [6], [22].

The rising energy consumption from EVs risks overloading grid capacity, potentially leading to instability and outages. To address these challenges, effective energy management solutions are critical, particularly through machine learning (ML) [22], [29], [30]. ML algorithms analyze historical charging patterns, energy consumption, and grid conditions to predict future demand [24], enabling strategies like optimizing energy distribution, reducing peak loads, and improving grid capacity [30]. Additionally, understanding driver behavior, including charging times and locations, helps refine ML models to improve grid operations and infrastructure planning [23].

In Bangladesh, limited localized EV data presents a challenge. To overcome this, our study uses California's EV charging data as a proxy [8], developing predictive models tailored to Bangladesh's needs. These models provide valuable insights into energy demands, assisting policymakers and grid operators in planning sustainable EV infrastructure while ensuring grid stability.

We do not claim this model as a solution to the energy crisis but as a tool to help anticipate and manage challenges from growing EV adoption. Our goal is to support the sustainable integration of EVs into Bangladesh's transportation system while ensuring grid resilience.

## 2 Related Work

The rapid growth of electric vehicles (EVs) has prompted significant research into optimizing charging infrastructure and predicting energy consumption patterns. Machine learning (ML) and deep learning techniques have been explored to improve demand forecasting, scheduling, and charging station efficiency [2, 4, 9, 10, 16, 19, 27, 28].

[10] analyzed energy consumption estimation, emphasizing adaptive models due to rapid developments in neural network architectures. The study focused on NeuralPower and SyNERGY for GPU-based estimation and fragmented ML ecosystems. [28] proposed an ML solution to optimize water pumping station operations on remote islands, achieving a 15% reduction in energy consumption and suggesting that integrating renewable energy sources could enhance efficiency further.



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[9] used multiple linear regression models to forecast residential energy consumption, improving accuracy by incorporating temperature and solar radiation, with potential for further refinement using smart meters. [27] combined Stationary Wavelet Transform (SWT) with deep transformers for household energy forecasting, achieving a 45% improvement in RMSE. [2] employed LSTM networks to estimate energy usage in educational facilities, achieving 94.31% accuracy. Future studies suggested adding device types to refine predictions. [19] reviewed hybrid and ensemble ML models for energy usage prediction, stressing the need for faster, more accurate models.

[4] explored how ML and AI optimize LoRaWAN performance by adjusting spreading factors and bandwidth, focusing on cross-layer optimization. [16] presented a hybrid model using CatBoost, XGBoost, and a multilayer perceptron for energy forecasting on Jeju Island, considering factors like weather and holidays.

The demand for EV infrastructure has driven research on optimizing charging stations and predicting energy use [1, 5, 11–14, 17]. [13] presented data from 22,200 charging stations in Germany, revealing trends in utilization and energy transfer. [1] used SARIMA models to estimate power usage at EV stations. [14] proposed a microgrid-based system to reduce operational costs at PV-powered stations.

Studies in [20–25, 29–32] investigated smart grids, centralized management, and scheduling algorithms to optimize EV charging, supporting EV infrastructure growth and minimizing grid strain. However, in Bangladesh, research on EV infrastructure remains limited, requiring tailored ML approaches to support sustainable EV growth.

### 3 Motivation

Bangladesh's energy infrastructure faces mounting pressure from the rising demand for EV charging stations. As EV adoption grows, efficient energy distribution and grid stability become critical. Factors like traffic patterns and peak-hour charging increase the unpredictability of demand, risking grid overload, outages, and inefficiencies. Accurate predictions of EV charging behavior are essential for resource optimization, preventing overloads, and ensuring smooth energy distribution. Without such insights, the infrastructure struggles to meet the EV market's needs sustainably.

This research addresses these challenges by developing machine learning-based predictive models to forecast EV charging behaviors and optimize energy distribution. Leveraging datasets from advanced EV markets, such as California, this study aims to create a scalable, adaptable framework suitable for Bangladesh's evolving EV infrastructure. These models enable proactive grid management, reduce environmental impacts, and guide sustainable EV infrastructure planning, laying the groundwork for future applications in developing countries.

### 4 Proposed Methodology

The methodology for predicting EV charging behavior involves analyzing key features such as session duration, energy consumption [22], gasoline savings, fees, and GHG reductions. Preprocessing steps include data cleaning, outlier detection, feature scaling, and encoding categorical variables. Exploratory Data Analysis (EDA)

identifies critical patterns, while feature engineering enhances the dataset for machine learning models.

Three models—Random Forest (RF), Gradient Boosting Regression (GBR), and Decision Tree (DT)—are utilized. RF reduces overfitting by averaging predictions across trees [3], GBR iteratively corrects errors for higher accuracy [7], and DT provides an interpretable structure for regression and classification tasks [3]. These models establish a robust framework for predicting EV charging behavior while effectively handling dataset complexities. At this stage, traditional models like RF and GBR provide a strong foundation, with the potential to adopt custom models or advanced techniques for Bangladesh's unique requirements.

To refine precision, K-fold cross-validation and hyperparameter optimization using Grid Search CV, Randomized Search CV, and HyperOpt [7] are employed. Performance metrics such as  $R^2$ , RMSE, and MAE ensure accurate energy demand forecasts, helping to manage grid capacity, highlight peak usage, and inform infrastructure planning.

## 5 Experimentation and Data Collection

The efficacy of a predictive machine learning model predominantly hinges on the quality of the dataset employed. In this research, the dataset used for modeling and forecasting EV charging behavior derives from the EV Charging Station Usage [8] records of California City. This dataset presents diverse and comprehensive insights, serving as the essential foundation for training and evaluating machine learning models designed to enhance the operational efficiency of EV charging stations.

### 5.1 Data Collection

The quality of the dataset is crucial for the success of predictive machine learning models. This study uses the California City EV Charging Station Usage dataset [8], which includes 259,416 charging sessions over several years. Key metrics cover station locations, connector types, session durations, start and end times, GHG and gasoline savings, and total energy consumption. These details offer valuable insights into typical EV charging behaviors, aiding in the optimization of station operations. Features such as charging duration, energy consumption, gasoline savings, and GHG reductions enhance behavior analysis, improving predictive accuracy, model efficiency, and operational performance.

### 5.2 Data Analysis

The analysis of EV charging events reveals rising demand from 2016 to 2019, followed by a decline in 2020 due to COVID-19. October sees peak activity, weekdays (Wednesday to Friday) show high usage, and weekends have lower demand, reflecting commuting patterns.

Figure 1 shows the total number of charge events across different years. The dataset spans from 2016 to 2020, with the highest volume of charge events recorded in 2017 (46,466), closely followed by 2019 (43,965) and 2018 (40,502). A marked decline in charge events appears in 2016 (20,177) and 2020 (18,336), likely attributable to early limitations in infrastructure and the effects of the COVID-19 pandemic, respectively.

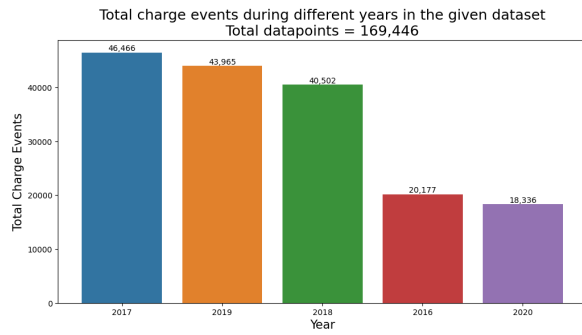


Figure 1: Total Charge Events by Year (2016-2020)

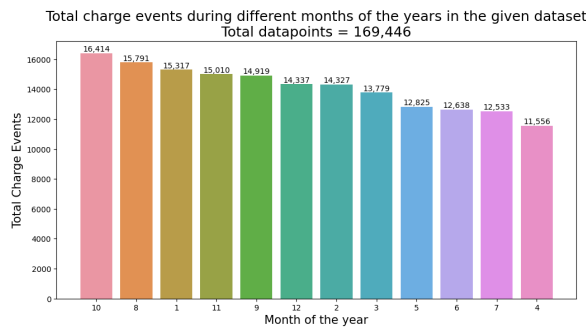


Figure 2: Total Charge Events by Month Across Multiple Years

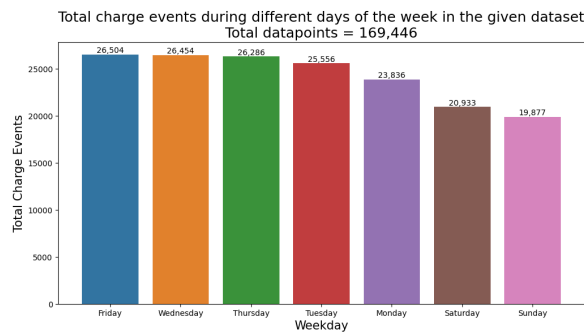


Figure 3: Total Charge Events by Weekday

Figure 2 illustrates the distribution of total charge events across different months of the year. October records the highest volume of charge events (16,414), followed by August (15,791) and November (15,317). The lowest volume is registered in April (11,556), which indicates potential seasonality in EV usage, with heightened demand during the fall and reduced activity in the spring months.

Figure 3 presents the total charge events by day of the week, with Friday (26,504), Wednesday (26,454), and Thursday (26,286) demonstrating the highest levels of activity. In contrast, weekends, particularly Sunday (19,877) and Saturday (20,933), exhibit a marked decline in charge events, suggesting that EV usage correlates with work commuting patterns, resulting in diminished demand over

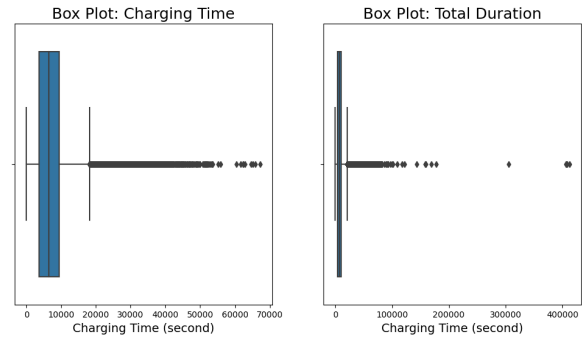


Figure 4: Box Plots of Charging Time and Total Duration

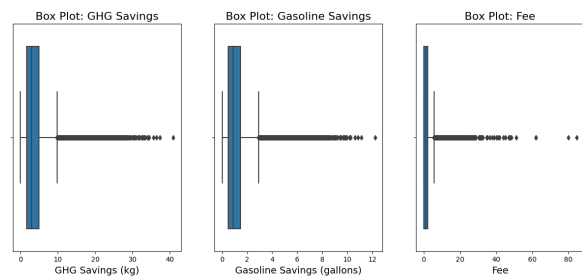


Figure 5: Box Plots of GHG Savings, Gasoline Savings, and Fees

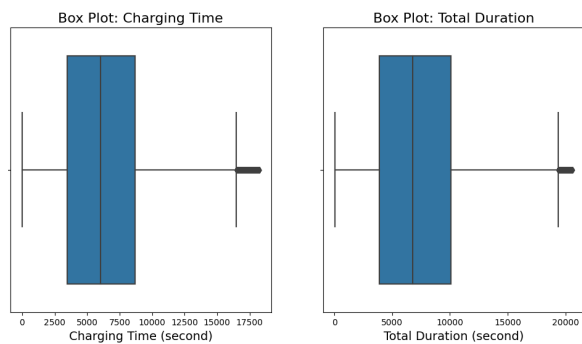


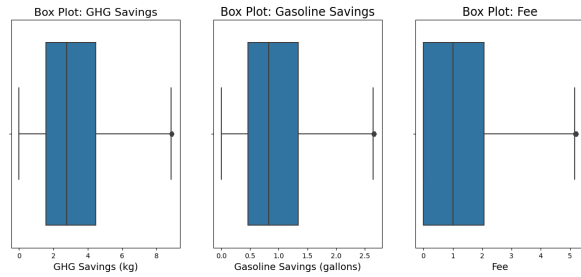
Figure 6: Box Plots of Charging Time and Total Duration (Refined View)

the weekends. These charging sessions likely contribute to the observed upward trend in power demand .

### 5.3 Data Preprocessing

Data cleaning [22] is crucial in preparing the dataset for model development. After removing null values, the dataset is reduced from 259,416 to 169,446 data points, covering July 2011 to December 2020. To further refine the data, the Interquartile Range (IQR) method is applied to identify and remove outliers, as they can negatively affect model performance. Boxplots are used to visualize potential outliers.

Figures 4 and 5 show outliers across all parameters, indicating variability in billing periods, session durations, savings, and costs.



**Figure 7: Box Plots of GHG Savings, Gasoline Savings, and Fees (Refined View)**

While most data points fall within a typical range, outliers may represent rare cases or errors. After removing outliers, Figures 6 and 7 reveal a more normalized distribution, highlighting core behavioral patterns. The central data emphasizes the medians for key parameters like charging time, total duration, GHG reductions, gasoline savings, and fees, offering a more accurate view.

#### 5.4 Feature Engineering

Feature engineering employs human expertise to convert raw data into a more understandable format, improving algorithm performance through creativity and domain knowledge [30]. Charging Time and Total Duration are converted from a human-readable format (hh:mm:ss) to seconds for easier analysis. A correlation heatmap [22] reveals strong positive correlations between energy usage (kWh) and overall duration (0.72), charging time (0.84), fuel savings (1), GHG savings (1), and charge (0.58). These five features are used to train the models, enhancing the prediction of energy consumption and related metrics.

#### 5.5 Model Selection and Analysis

We analyze EV charging behavior using California City’s EV Charging Station Usage [8] records from July 2011 to December 2020. The dataset is split into 80% for training and 20% for testing. We train Random Forest, Gradient Boosting Regression, and Decision Tree models [3], [7]. K-fold cross-validation [24] is applied during training, where the dataset is randomly partitioned into K equal pieces, and each model is trained K times. The Decision Tree and Random Forest use K=10, while Gradient Boosting Regression uses K=2, ensuring multiple tests on different data subsets.

Table 1 presents the initial hyperparameters for Grid Search CV [24], Randomized Search CV, and HyperOpt with cross-validation [26]. Grid Search CV evaluates all hyperparameter combinations, while Randomized Search CV tests a random subset for efficiency. HyperOpt uses Bayesian optimization to focus on promising areas of the hyperparameter space. All three methods employ cross-validation to enhance model performance and accuracy.

### 6 Results and Analysis

Table 2 summarizes the hyperparameters and their optimized values obtained via Grid Search CV, Randomized Search CV, and HyperOpt using K-fold validation for Random Forest, Decision Tree, and Gradient Boosting Regression. Gradient Boosting Regression allows

**Table 1: Hyperparameters for different ML Algorithms**

| ML Algorithms               | Hyperparameter    | List of Value               |
|-----------------------------|-------------------|-----------------------------|
| Random Forest (RF)          | n_estimators      | [100, 200, 300]             |
|                             | max_features      | ['sqrt', 'log2']            |
|                             | max_depth         | [10, 20, 30]                |
|                             | min_samples_split | [2, 5, 10]                  |
|                             | min_samples_leaf  | [1, 2, 4]                   |
| Gradient Boosting Regressor | max_depth         | [5, 10, 15]                 |
|                             | min_samples_split | [10, 20, 30]                |
|                             | min_samples_leaf  | [5, 10, 20]                 |
|                             | max_leaf_nodes    | [50, 100, 150]              |
| Decision Tree               | n_estimators      | [100, 300, 10]              |
|                             | learning rate     | [np.log(0.01), np.log(0.1)] |
|                             | max_depth         | [2, 6, 1]                   |
|                             | min_sample_split  | [10, 30, 5]                 |
|                             | min_samples_leaf  | [5, 15, 5]                  |
|                             | subsample         | [0.7, 1.0]                  |
|                             | max_features      | ['sqrt', 'log2']            |

greater customization, including learning rate and subsample, while Random Forest and Decision Tree focus on core parameters like tree depth, splitting criteria, and the number of trees or leaf nodes. These parameters are crucial for controlling model complexity, avoiding overfitting, and enhancing predictive accuracy.

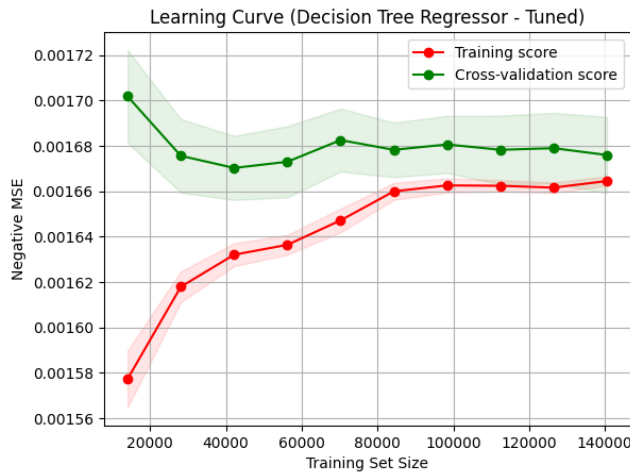
**Table 2: Tuning Parameters Obtained from Grid Search, Randomized Search CV and HyperOpt with K-folds Cross-Validation**

| ML Algorithms               | Hyperparameter    | List of Value |
|-----------------------------|-------------------|---------------|
| Random Forest               | n_estimators      | [100]         |
|                             | max_features      | ['sqrt']      |
|                             | max_depth         | [20]          |
|                             | min_samples_split | [5]           |
|                             | min_samples_leaf  | [1]           |
| Gradient Boosting Regressor | max_depth         | [15]          |
|                             | min_samples_split | [20]          |
|                             | min_samples_leaf  | [5]           |
|                             | max_leaf_nodes    | [150]         |
| Decision Tree               | n_estimators      | [220]         |
|                             | learning rate     | [0.04158]     |
|                             | max_depth         | [6]           |
|                             | min_samples_split | [10]          |
|                             | min_samples_leaf  | [10]          |
|                             | subsample         | [0.934]       |
|                             | max_features      | [0]           |

Table 3 compares the performance of Random Forest, Decision Tree, and Gradient Boosting with R2, RMSE, and MAE [24] measures. Random Forest obtains an R2 of 0.9999944, while Gradient Boosting has a slightly higher R2 of 0.9999965. The Decision Tree falls behind with an R2 of 0.9999272. All models have remarkable R2 values. Random Forest has the lowest RMSE at 0.003, but a higher MAE of 0.0874, implying worse precision. The decision tree has a

**Table 3: EVALUATION METRICS OF ALL MODELS**

| ML Algorithms               | R <sup>2</sup> | RMSE   | MAE    |
|-----------------------------|----------------|--------|--------|
| Random Forest               | 0.9999944      | 0.0035 | 0.0874 |
| Gradient Boosting Regressor | 0.9999965      | 0.0088 | 0.0071 |
| Decision Tree               | 0.9999272      | 0.0410 | 0.0342 |

**Figure 8: Learning Curve (Decision Tree)**

higher RMSE of 0.0410 and MAE of 0.0342. In comparison, Gradient Boosting has the lowest MAE (0.0071), making it the most exact model.

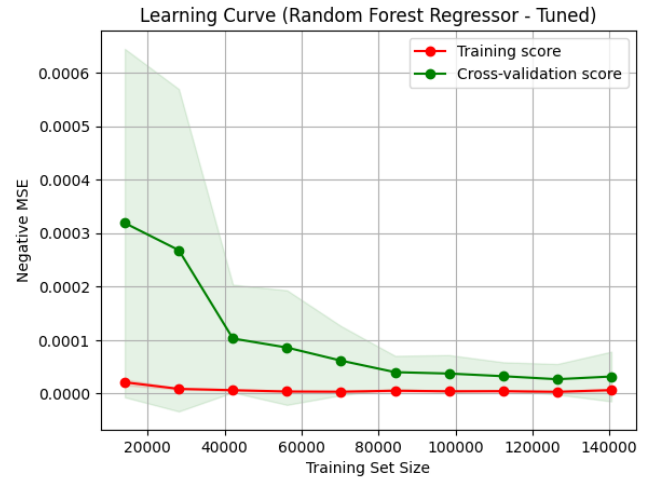
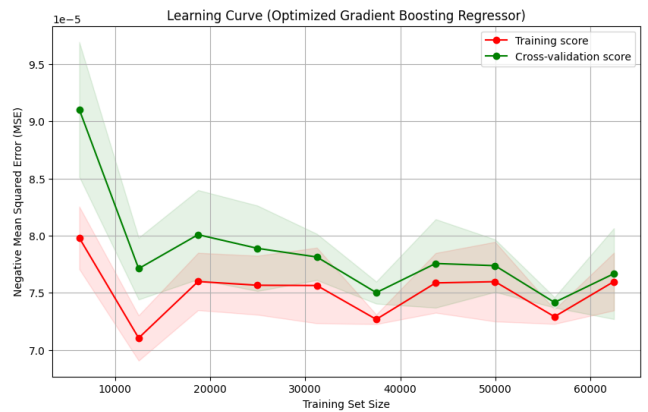
### 6.1 Learning Curve Analysis for Decision Tree

The learning curve for the tuned Decision Tree Regression, as shown in Figure 8, the red line indicates that the model accurately matches the training data, with a very low negative mean squared error(MSE).

The green line indicates a larger error in the cross-validation set compared to the training set, with a significant difference between the two scores. This signals overfitting, in which the model performs well on training data but does not generalize well to new data. The gap suggests that the model captures training data trends but struggles with unseen data.

### 6.2 Learning Curve Analysis for Random Forest

As depicted in Figure 9, The training error is consistently low and flat across all training sizes, showing that the model fits the training data perfectly. The cross-validation score starts high and decreases as the training set size increases, getting closer to the training error but still leaving a tiny gap. This model has a minor overfitting, however as the training set size grows, the cross-validation error approaches the training error. The diminishing gap shows that when more data is collected, the models' generalization will improve. The variance between training and validation scores diminishes dramatically when more data is added.

**Figure 9: Learning Curve (Random Forest)****Figure 10: Learning Curve (Gradient Boosting Regressor)**

### 6.3 Learning Curve Analysis for Gradient Boosting Regressor

As shown in Figure 10, the training error initially decreases and then stabilizes at a low level, indicating good model fit. The cross-validation error also decreases but stabilizes at a higher level than the training error, with some fluctuations as the training set grows. This suggests modest overfitting, as the cross-validation error exceeds the training error. However, the gap is smaller compared to the Decision Tree, showing better generalization with room for improvement in variance.

In contrast, the Decision Tree Regressor and Random Forest Regressor exhibit greater overfitting, especially the Decision Tree, where the gap between training and cross-validation errors is more pronounced. None of the models indicate underfitting, as both errors are relatively low. Notably, increasing the training set size improves generalization across all models. The Gradient Boosting

Regressor demonstrates the best balance, with consistent performance, reduced fluctuations, and a smaller error gap, making it the most well-rounded model.

## 7 Conclusion

This study addresses the critical need to estimate energy consumption at EV charging stations in Bangladesh, where rapid EV adoption increasingly strains the national power grid. Employing machine learning models such as Random Forest, Gradient Boosting Regression, and Decision Tree, the study identifies Gradient Boosting as the most accurate predictor, enabling optimized energy distribution, mitigating grid overload [30], and improving resource utilization. In the absence of localized data, the study utilizes the California City dataset as a proxy [8], offering insights into future energy demands and supporting strategic infrastructure planning. As EV growth intensifies challenges for the already overburdened grid [18], this research provides a foundation for balancing energy supply and demand, advancing grid stability, and fostering long-term sustainability. Addressing a critical gap in the literature, it establishes a proactive framework for policymakers and grid operators, with future efforts focusing on integrating real-time, localized data to further refine predictions and support the transition to cleaner transportation systems

## References

- [1] K. Akshay, Hannah Grace, KANIMOZHI G, and Ravi Samikannu. 2024. Power consumption prediction for electric vehicle charging stations and forecasting income. *Scientific Reports* 14 (03 2024). <https://doi.org/10.1038/s41598-024-56507-2>
- [2] Maha Alanbar, Amal Alfarraj, and Manal Alghith. 2020. Energy Consumption Prediction Using Deep Learning Technique. *International Journal of Interactive Mobile Technologies (IJIM)* 14, 10 (Jun. 2020), pp. 166–177. <https://doi.org/10.3991/ijim.v14i10.14383>
- [3] Jehad Ali, Rehanullah Khan, Nasir Ahmad, and Imran Maqsood. 2012. Random forests and decision trees. *International Journal of Computer Science Issues (IJCSI)* 9, 5 (2012), 272.
- [4] Maram Alkhaayal and Almetwally Mostafa. 2024. Recent Developments in AI and ML for IoT: A Systematic Literature Review on LoRaWAN Energy Efficiency and Performance Optimization. *Sensors* 24, 14 (2024). <https://doi.org/10.3390/s24144482>
- [5] Youssef Amry, Elhoussin Elbouchikhi, Franck Le Gall, Mounir Ghogho, and Soumia El Hani. 2022. Electric Vehicle Traction Drives and Charging Station Power Electronics: Current Status and Challenges. *Energies* 15, 16 (2022). <https://doi.org/10.3390/en15166037>
- [6] Anirban Barua, Abdullah Al Mahbub, Mazbah Kabir Mridul, Sazzad Ahmad, Md Harunur Rashid, and Partha Mandal. 2022. Designing SWET Based Hybrid Charging Stations for Electric Vehicles at Dhaka-Mawa Expressway in Bangladesh. In *2022 4th International Conference on Sustainable Technologies for Industry 4.0 (STI)*. 1–6. <https://doi.org/10.1109/STI56238.2022.10103336>
- [7] Candice Bentéjac, Anna Csörgő, and Gonzalo Martínez-Muñoz. 2021. A comparative analysis of gradient boosting algorithms. *Artificial Intelligence Review* 54 (2021), 1937–1967.
- [8] Venkatasai Rao Dheekonda. [n. d.]. *EV Charging Station Usage of California City*. <https://www.kaggle.com/datasets/venkatsai4899/ev-charging-station-usage-of-california-city>
- [9] Nelson Fumo and M.A. Rafe Biswas. 2015. Regression analysis for prediction of residential energy consumption. *Renewable and Sustainable Energy Reviews* 47 (2015), 332–343. <https://doi.org/10.1016/j.rser.2015.03.035>
- [10] Eva García-Martín, Crefeda Faviola Rodrigues, Graham Riley, and Håkan Grahm. 2019. Estimation of energy consumption in machine learning. *J. Parallel and Distrib. Comput.* 134 (2019), 75–88. <https://doi.org/10.1016/j.jpdc.2019.07.007>
- [11] Madeline Gilleran, Eric Bonnema, Jason Woods, Partha Mishra, Ian Doebber, Chad Hunter, Matt Mitchell, and Margaret Mann. 2021. Impact of electric vehicle charging on the power demand of retail buildings. *Advances in Applied Energy* 4 (2021), 100062. <https://doi.org/10.1016/j.adapen.2021.100062>
- [12] Abdul Hafeez, Rashid Alammar, and Atif Iqbal. 2023. Utilization of EV Charging Station in Demand Side Management Using Deep Learning Method. *IEEE Access* 11 (2023), 8747–8760. <https://doi.org/10.1109/ACCESS.2023.3238667>
- [13] Christopher Hecht, Jan Figgner, and Dirk Uwe Sauer. 2022. Analysis of electric vehicle charging station usage and profitability in Germany based on empirical data. *iScience* 25, 12 (2022), 105634. <https://doi.org/10.1016/j.isci.2022.105634>
- [14] Hojun Jin, Sangkeum Lee, Sarvar Hussain Nengroo, and Dongsoo Har. 2022. Development of Charging/Discharging Scheduling Algorithm for Economical and Energy-Efficient Operation of Multi-EV Charging Station. *Applied Sciences* 12, 9 (2022). <https://doi.org/10.3390/app12094786>
- [15] Ashish Karmaker. 2019. Challenges for Electric Vehicle Adoption in Bangladesh. <https://doi.org/10.1109/ECACE.2019.8679288>
- [16] Prince Waqas Khan, Yongjun Kim, Yung-Cheol Byun, and Sang-Joon Lee. 2021. Influencing Factors Evaluation of Machine Learning-Based Energy Consumption Prediction. *Energies* 14, 21 (2021). <https://doi.org/10.3390/en14217167>
- [17] Kai Li Lim, Stuart Speidel, and Thomas Bräunl. 2022. A comparative study of AC and DC public electric vehicle charging station usage in Western Australia. *Renewable and Sustainable Energy Transition* 2 (2022), 100021. <https://doi.org/10.1016/j.rset.2022.100021>
- [18] Tehseen Mazhar, Rizwana Naz Asif, Muhammad Amir Malik, Muhammad Asgher Nadeem, Inayatul Haq, Muhammad Iqbal, Muhammad Kamran, and Shahzad Ashraf. 2023. Electric Vehicle Charging System in the Smart Grid Using Different Machine Learning Methods. *Sustainability* 15, 3 (2023). <https://doi.org/10.3390/su15032603>
- [19] Amir Mosavi and Abdullah Bahmani. 2019. Energy Consumption Prediction Using Machine Learning; A Review. (03 2019).
- [20] Boulakbhar Mouaad, Markos Farag, Kawtar Benabdelaziz, Tarik Kousksou, and Malika Zazi. 2022. A Deep Learning Approach for Prediction of Electrical Vehicle Charging Stations Power Demand in Regulated Electricity Markets: The Case of Morocco. *Cleaner Energy Systems* 3 (11 2022), 100039. <https://doi.org/10.1016/j.cles.2022.100039>
- [21] Fenil Ramoliya, Chinmay Trivedi, Krisha Darji, Riya Kakkar, Rajesh Gupta, Sudeep Tanwar, Zdzislaw Polkowski, Faye Alqahtani, and Amr Tolba. 2024. ML-Based Energy Consumption and Distribution Framework Analysis for EVs and Charging Stations in Smart Grid Environment. *IEEE Access* 12 (2024), 23319–23337. <https://doi.org/10.1109/ACCESS.2024.3365080>
- [22] Harshit Rathore, Hemant Kumar Meena, and Prerna Jain. 2023. Forecasting of EVs Charging Behavior Using Deep Neural Networks. In *2023 International Conference on Communication, Circuits, and Systems (ICCS)*. 1–6. <https://doi.org/10.1109/ICCS57698.2023.10169211>
- [23] Harshit Rathore, Hemant Kumar Meena, and Prerna Jain. 2023. Prediction of EV Energy consumption Using Random Forest And XGBoost. In *2023 International Conference on Power Electronics and Energy (ICPEE)*. 1–6. <https://doi.org/10.1109/ICPEE54198.2023.10060798>
- [24] Dionysius A. Renata, Khotimatul Fauziah, Prasetyo Aji, Adisa Larasati, Hafsa Halidah, Dwidharma Priyasta Tasurun, Yuli Astriani, and Riza. 2021. Modeling of Electric Vehicle Charging Energy Consumption using Machine Learning. In *2021 International Conference on Advanced Computer Science and Information Systems (ICACSIS)*. 1–6. <https://doi.org/10.1109/ICACSIS53237.2021.9631324>
- [25] Dionysius A. Renata, Khotimatul Fauziah, Prasetyo Aji, Adisa Larasati, Hafsa Halidah, Dwidharma Priyasta Tasurun, Yuli Astriani, and Riza. 2021. Modeling of Electric Vehicle Charging Energy Consumption using Machine Learning. In *2021 International Conference on Advanced Computer Science and Information Systems (ICACSIS)*. 1–6. <https://doi.org/10.1109/ICACSIS53237.2021.9631324>
- [26] Yagyanath Rimal, Navneet Sharma, and Abeer Alsadoon. 2024. The accuracy of machine learning models relies on hyperparameter tuning: student result classification using random forest, randomized search, grid search, bayesian, genetic, and optuna algorithms. *Multimedia Tools and Applications* (2024), 1–16.
- [27] Lyes Saad Saoud, Hasan Al-Marzouqi, and Ramy Hussein. 2022. Household Energy Consumption Prediction Using the Stationary Wavelet Transform and Transformers. *IEEE Access* 10 (2022), 5171–5183. <https://doi.org/10.1109/ACCESS.2022.3140818>
- [28] Elissaios Sarmas, Evangelos Spiliotis, Vangelis Marinakis, Georgios Tzanes, John Kaldellis, and H. Doukas. 2022. ML-based energy management of water pumping systems for the application of peak shaving in small-scale islands. *Sustainable Cities and Society* 82 (04 2022), 103873. <https://doi.org/10.1016/j.scs.2022.103873>
- [29] Sakib Shahriar, A. R. Al-Ali, Ahmed H. Osman, Salam Dhou, and Mais Nijim. 2020. Machine Learning Approaches for EV Charging Behavior: A Review. *IEEE Access* 8 (2020), 168980–168993. <https://doi.org/10.1109/ACCESS.2020.3023388>
- [30] Sakib Shahriar, A. R. Al-Ali, Ahmed H. Osman, Salam Dhou, and Mais Nijim. 2021. Prediction of EV Charging Behavior Using Machine Learning. *IEEE Access* 9 (2021), 111576–111586. <https://doi.org/10.1109/ACCESS.2021.3103119>
- [31] Irfan Ullah, Kai Liu, Toshiyuki Yamamoto, Muhammad Zahid, and Arshad Jamal. 2023. Modeling of machine learning with SHAP approach for electric vehicle charging station choice behavior prediction. *Travel Behaviour and Society* 31 (2023), 78–92. <https://doi.org/10.1016/j.tbs.2022.11.006>
- [32] Qingbo Zhu, Yicun Huang, Chih Feng Lee, Peng Liu, Jin Zhang, and Torsten Wik. 2024. Predicting Electric Vehicle Energy Consumption from Field Data Using Machine Learning. *IEEE Transactions on Transportation Electrification* (2024), 1–1. <https://doi.org/10.1109/TTE.2024.3416532>